Conversions between standard polynomial bases

Terence Tao · 7 April 2019

https://terrytao.wordpress.com/2019/04/07/conversions-between-standard-polynomial-bases/

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(This post is mostly intended for my own reference, as I found myself repeatedly looking up several conversions between polynomial bases on various occasions.)

Let $\mathrm{Poly}_{\leq n}$ denote the vector space of polynomials $P : \mathbf{R} \to \mathbf{R}$ of one variable x with real coefficients of degree at most n. This is a vector space of dimension $n+1$, and the sequence of these spaces form a filtration:

$$\mathrm{Poly}_{\leq 0} \subset \mathrm{Poly}_{\leq 1} \subset \mathrm{Poly}_{\leq 2} \subset \ldots$$

A standard basis for these vector spaces are given by the monomials $x^0, x^1, x^2, \ldots$: every polynomial $P(x)$ in $\mathrm{Poly}_{\leq n}$ can be expressed uniquely as a linear combination of the first $n+1$ monomials $x^0, x^1, \ldots, x^n$. More generally, if one has any sequence $Q_0(x), Q_1(x), Q_2(x)$ of polynomials, with each Q_n of degree exactly n, then an easy induction shows that $Q_0(x), \ldots, Q_n(x)$ forms a basis for $\mathrm{Poly}_{\leq n}$.

In particular, if we have *two* such sequences $Q_0(x), Q_1(x), Q_2(x), \ldots$ and $R_0(x), R_1(x), R_2(x), \ldots$ of polynomials, with each Q_n of degree n and each R_k of degree k, then Q_n must be expressible uniquely as a linear combination of the polynomials $R_0, R_1, \ldots, R_n$, thus we have an identity of the form

$$Q_n(x) = \sum_{k=0}^{n} c_{QR}(n,k) R_k(x)$$

for some *change of basis coefficients* $c_{QR}(n,k) \in \mathbf{R}$. These coefficients describe how to convert a polynomial expressed in the Q_n basis into a polynomial expressed in the R_k basis.

Many standard combinatorial quantities $c(n,k)$ involving two natural numbers $0 \leq k \leq n$ can be interpreted as such change of basis coefficients. The most familiar example are the binomial coefficients $\binom{n}{k}$, which measures the conversion from the shifted monomial basis $(x+1)^n$ to the monomial basis x^k, thanks to (a special case of) the binomial formula:

$$(x+1)^n = \sum_{k=0}^{n} \binom{n}{k} x^k,$$

thus for instance

$$\begin{aligned}(x+1)^3 &= \binom{3}{0}x^0 + \binom{3}{1}x^1 + \binom{3}{2}x^2 + \binom{3}{3}x^3 \\ &= 1 + 3x + 3x^2 + x^3.\end{aligned}$$

More generally, for any shift h, the conversion from $(x+h)^n$ to x^k is measured by the coefficients $h^{n-k}\binom{n}{k}$, thanks to the general case of the binomial formula.

But there are other bases of interest too. For instance if one uses the falling factorial basis

$$(x)_n := x(x-1)\ldots(x-n+1)$$

then the conversion from falling factorials to monomials is given by the Stirling numbers of the first kind $s(n,k)$:

$$(x)_n = \sum_{k=0}^{n} s(n,k) x^k,$$

thus for instance

$$\begin{aligned}(x)_3 &= s(3,0)x^0 + s(3,1)x^1 + s(3,2)x^2 + s(3,3)x^3 \\ &= 0 + 2x - 3x^2 + x^3\end{aligned}$$

and the conversion back is given by the Stirling numbers of the second kind $S(n,k)$:

$$x^n = \sum_{k=0}^{n} S(n,k)(x)_k$$

thus for instance

$$\begin{aligned}x^3 &= S(3,0)(x)_0 + S(3,1)(x)_1 + S(3,2)(x)_2 + S(3,3)(x)_3 \\ &= 0 + x + 3x(x-1) + x(x-1)(x-2).\end{aligned}$$

If one uses the binomial functions $\binom{x}{n} = \frac{1}{n!}(x)_n$ as a basis instead of the falling factorials, one of course can rewrite these conversions as

$$\binom{x}{n} = \sum_{k=0}^{n} \frac{1}{n!} s(n,k) x^k$$

and

$$x^n = \sum_{k=0}^{n} k! S(n,k) \binom{x}{k}$$

thus for instance

$$\binom{x}{3} = 0 + \frac{1}{3}x - \frac{1}{2}x^2 + \frac{1}{6}x^3$$

and

$$x^3 = 0 + \binom{x}{1} + 6\binom{x}{2} + 6\binom{x}{3}.$$

As a slight variant, if one instead uses rising factorials

$$(x)^n := x(x+1)...(x+n-1)$$

then the conversion to monomials yields the unsigned Stirling numbers $|s(n,k)|$ of the first kind:

$$(x)^n = \sum_{k=0}^{n} |\ s(n,k)|\ x^k$$

thus for instance

$$(x)^3 = 0 + 2x + 3x^2 + x^3.$$

One final basis comes from the polylogarithm functions

$$\mathrm{Li}_{-n}(x) := \sum_{j=1}^{\infty} j^n x^j.$$

For instance one has

$$\begin{aligned}
\mathrm{Li}_1(x) &= -\log(1-x) \\
\mathrm{Li}_0(x) &= \frac{x}{1-x} \\
\mathrm{Li}_{-1}(x) &= \frac{x}{(1-x)^2} \\
\mathrm{Li}_{-2}(x) &= \frac{x}{(1-x)^3}(1+x) \\
\mathrm{Li}_{-3}(x) &= \frac{x}{(1-x)^4}(1+4x+x^2) \\
\mathrm{Li}_{-4}(x) &= \frac{x}{(1-x)^5}(1+11x+11x^2+x^3)
\end{aligned}$$

and more generally one has

$$\mathrm{Li}_{-n-1}(x) = \frac{x}{(1-x)^{n+2}} E_n(x)$$

for all natural numbers n and some polynomial E_n of degree n (the *Eulerian polynomials*), which when converted to the monomial basis yields the (shifted) Eulerian numbers

$$E_n(x) = \sum_{k=0}^{n} A(n+1,k) x^k.$$

For instance

$$\begin{aligned}
E_3(x) &= A(4,0)x^0 + A(4,1)x^1 + A(4,2)x^2 + A(4,3)x^3 \\
&= 1 + 11x + 11x^2 + x^3.
\end{aligned}$$

These particular coefficients also have useful combinatorial interpretations. For instance:

- The binomial coefficient $\binom{n}{k}$ is of course the number of k-element subsets of $\{1, \dots, n\}$.
- The unsigned Stirling numbers $|s(n,k)|$ of the first kind are the number of permutations of $\{1, \dots, n\}$ with exactly k cycles. The signed Stirling numbers $s(n,k)$ are then given by the formula $s(n,k) = (-1)^{n-k}|s(n,k)|$.
- The Stirling numbers $S(n,k)$ of the second kind are the number of ways to partition $\{1, \dots, n\}$ into k non-empty subsets.
- The Eulerian numbers $A(n,k)$ are the number of permutations of $\{1, \dots, n\}$ with exactly k ascents.

These coefficients behave similarly to each other in several ways. For instance, the binomial coefficients $\binom{n}{k}$ obey the well known Pascal identity

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

(with the convention that $\binom{n}{k}$ vanishes outside of the range $0 \leq k \leq n$). In a similar spirit, the unsigned Stirling numbers $|s(n,k)|$ of the first kind obey the identity

$$|s(n+1,k)| = n|s(n,k)| + |s(n,k-1)|$$

and the signed counterparts $s(n,k)$ obey the identity

$$s(n+1,k) = -ns(n,k) + s(n,k-1).$$

The Stirling numbers of the second kind $S(n,k)$ obey the identity

$$S(n+1,k) = kS(n,k) + S(n,k-1)$$

and the Eulerian numbers $A(n,k)$ obey the identity

$$A(n+1,k) = (k+1)A(n,k) + (n-k+1)A(n,k-1).$$

Comments (24)

Per Ekman · 7 April 2019

As someone that's currently learning a bout the Binomial Theorem on my free time this couldn't have been posted at a better time.
Thank you so much for sharing!

asahay22 · 7 April 2019

Hmmm, do the basis change coefficients from the standard basis to an orthogonal basis relative to some integral inner product (e.g. Chebyshev polynomials) also have a combinatorial interpretation?

> **Terence Tao** · 7 April 2019
>
> The basis change coefficients between the classical orthogonal polynomials and the monomial basis tend to be variants of binomial coefficients (for instance they often are expressible as ratios of products of factorials, times some other simple factors like powers of 2). For instance to convert monomials x^n to Chebyshev polynomials $T_k(x)$, one has the identity $x^n = \sum_{k=0}^n 2^{1_{k\neq 0}-n} \binom{n}{(n-k)/2} T_k(x)$ (with the convention that the binomial coefficient vanishes if the bottom argument is not an integer).
>
> > **Jake** · 8 April 2019
> >
> > See also things like Salzer's method : https://dl.acm.org/citation.cfm?doid=355611.362548 and this chebfun discussion : http://www.chebfun.org/examples/cheb/FastChebyshevLegendreTransform.html
>
> > **Anonymous** · 8 April 2019
> >
> > The coefficients of (weighted) orthogonal polynomials expansion can also be evaluated by their corresponding (weighted) integral representation as inner products.

Lou · 7 April 2019

Very nice. Any analogues for multivariate polynomials?

> **Anonymous** · 11 April 2019
>
> See e.g. spherical harmonics (which are orthogonal polynomials on the unit sphere S^2, or the more general S^n with respect to its Haar measure) with application to polynomial approximation on spheres.

Pedro Sánchez Terraf · 8 April 2019

Dear Terry,
in the second line of the post you might like "...one variable x and of degree at most n".
Best wishes!
[Corrected, thanks – T.]

greghighermathhelp · 8 April 2019

I was just learning about the calculus of finite differences last week, where the conversion of monomials into falling factorials allows for the evaluation of sums of kth powers, which I learned is a problem with a very long history. Now here it is again, with a nice connection to binomial coefficients and other combinatorial quantities! Thanks for fleshing this out.

Conversions between standard polynomial bases - Nevin Manimala's Blog · 9 April 2019
[...] by /u/HigherMathHelp [link] [...]

LaurentC · 9 April 2019

There may lack a “minus” in your formula of $Li_1=-\long(1-x)$.
[Corrected, thanks – T.]

Arch1 · 10 April 2019

2nd paragraph of the post proper: “…can be expressed uniquely as *a linear combination of* the first n+1 monomials…”
[Corrected, thanks – T.]

Raymond E Rogers · 11 April 2019

You can systematize a lot of polynomial conversions and sequences by means of Sheffer generating functions and substituting an appropriate “creation” matrix for the indexing variable; giving a coefficient matrix. The Appell sequences are straightforward, and the generalized forms (Sheffer) require going through the matrix Jordan form.

Rodolfo Niborski · 11 April 2019

Three formulas from the bottom :
“and the signed counterparts” instead of “unsigned”.
[Corrected, thanks – T.]

Ralph Burger · 12 April 2019

I once tried to find an explicit transformation for Fibonacci polynomials from $F_n(x+1)$ to the $F_m(x)$ basis, but failed to get a recursion free expression. Any ideas on that?

> **Terence Tao** · 12 April 2019
>
> The first thing I would suggest is to compute the first few coefficients and enter that into the OEIS to see if you get a hit. (For instance, the Stirling numbers of the first kind appear as https://oeis.org/A008275 , the Stirling numbers of the second kind appear as https://oeis.org/A008277 , and the Eulerian numbers appear as https://oeis.org/A008292 .) If the sequence is not already in the OEIS, you can of course submit it in case some future researcher also stumbles upon it.
> As an alternative to the OEIS one can also simply do a web search for some of the specific large numerical coefficients that you find. (For instance, if you pick a handful of large unsigned Stirling numbers of the first kind, such as 109584, 22449, and 13132, and do a web search, many of the hits will relate to the Stirling numbers.)
>
> > **Raphael Berger** · 12 April 2019
> >
> > Thanks a lot for replying (my grandchildren will hear that)! OEIS is a great suggestion; I did it, just didn’t mention it, thought that won’t fit the small margin here (https://math.stackexchange.com/questions/2731420/variable-transformation-x-to-x1-in-fibonacci-polynomial-f-nx) ;-) So its not there, and afaiu 2D sequences are not really supported by OIES. (Since we already meet, I have a thought on the Töplitz conjecture, you don’t happen to have a post on it here?). All the best!! — Raphael B.
>
> **Raymond E Rogers** · 12 April 2019
>
> I might not understand but the recursion in these cases (rational polynomial generating functions) is always the denominator applied to the series, with the numerator acting as startup/ arbitrary constants.

Ralph Burger · 12 April 2019

I do not understand exactly what you mean by apply, but as I wrote, the recursion is the easy bit, I aimed at the recursion-FREE expression (which might be a bad literal translation from the german “rekursionsfrei” and could be possibly paraphrased: “non-recursive”).

Raymond E Rogers · 12 April 2019

As I read it, you want the closed form of the generating function? In that case, just substitute your x+1 into the generating function and do a partial fraction expansion on it. Then solve the first order terms in closed form. We should probably move this discussion out of this forum.

Raymond Rogers · 12 April 2019

It occurs to me that you just want new coefficients? In an elementary form: just write out the polynomial as a coefficient array times a verticle base array; say (C_(x+1))*B1(X+1) Now expand (x+1)^n and you see it’s just a summation of binomial entries times X. Treat it as a convolution (or some such) and you see that you can convert the base to P()*B2(X) Where P() is the Pascal matrix with terms (n,k). Having rewritten it, you just use “associativity” and the new polynomial coefficients are C_x=C_(x+1)*P() and the new polynomial (actually the same one) reads C_x*B2(x). I hope this is closer to what you want.

Anonymous · 13 April 2019

Dear Raymond, thank you for your suggestion. My email is berger@chem.helsinki.fi. Maybe its better to continue
by email.

Anonymous · 13 April 2019

ps. in the link to math stack exchange you seeexactly how far i got. the problem is the secobd step after the binom. expansion to get back to fibonacci polynomials in x.

JDM · 15 April 2019

I’m going to confess that I thought a lot of the discussion of Hilbert spaces of Lp spaces could be ameliorated if we use a basis of some kind. Has anyone in number theory computed a Gowers norm yet?